Objective of the LAB.

To deploy “http” service on 2 EC2 instance from CodeDeploy and also copy an test page to the EC2 instance.

Result:

We should have an web service running on the EC2 and we should be able to access it via the EC2 instance public IP.

Lab-Setups – CodeDeploy

1. Create IAM Policy & Role for EC2 to access S3

    1. Provide – S3 – get and list Permission for the EC2 instance. -- EC2-codedeploy

    Example for the name , You can have same name for Role and Policy,eg: -- EC2-codedeploy

2. Create Ec2 instance with “EC2-codedeploy” Roles attached to it

3. Upload the “Sample-linuz.zip” into S3 bucket and give “Public” access

4. Install the Codedeploy agent on the EC2 instance.

5. Create IAM Policy & Role for CodeDeploy to Access EC2 instance

    Create Roles ,eg: -- and CodeDeploy

6. Create an CodeDeploy deployment code.

7. Create an ‘New Deploy” and point to the S3 location of the Sample-linux.zip file

8. Update the index.html file in the “.zip” file, and change “Congratulations” Edit the “appsepc” file. Remove all the lines from “hooks” and save it.

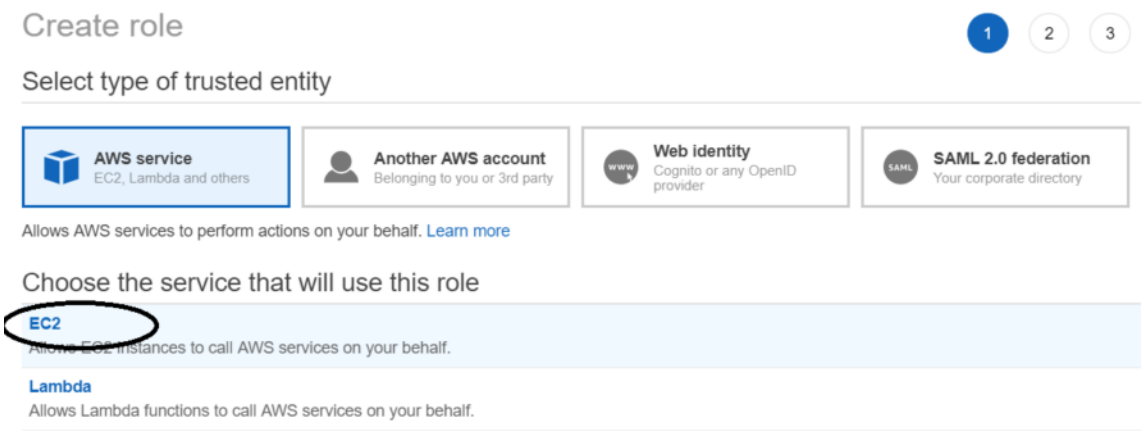
9. Then zip the files and upload it as diff version number example, “Sample-linux-v2.zip” ,

Step 1. Policy Creation

a. **IAM Policy** -- **EC2-codedeploy** for EC2 to access S3

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Action": [
                "s3:Get*",
                "s3:List*"
            ],
            "Effect": "Allow",
            "Resource": "*"
        }
    ]
}
```

b. IAM Role for **EC2-codedeploy**



Then attach the policy "**EC2-codedeploy**

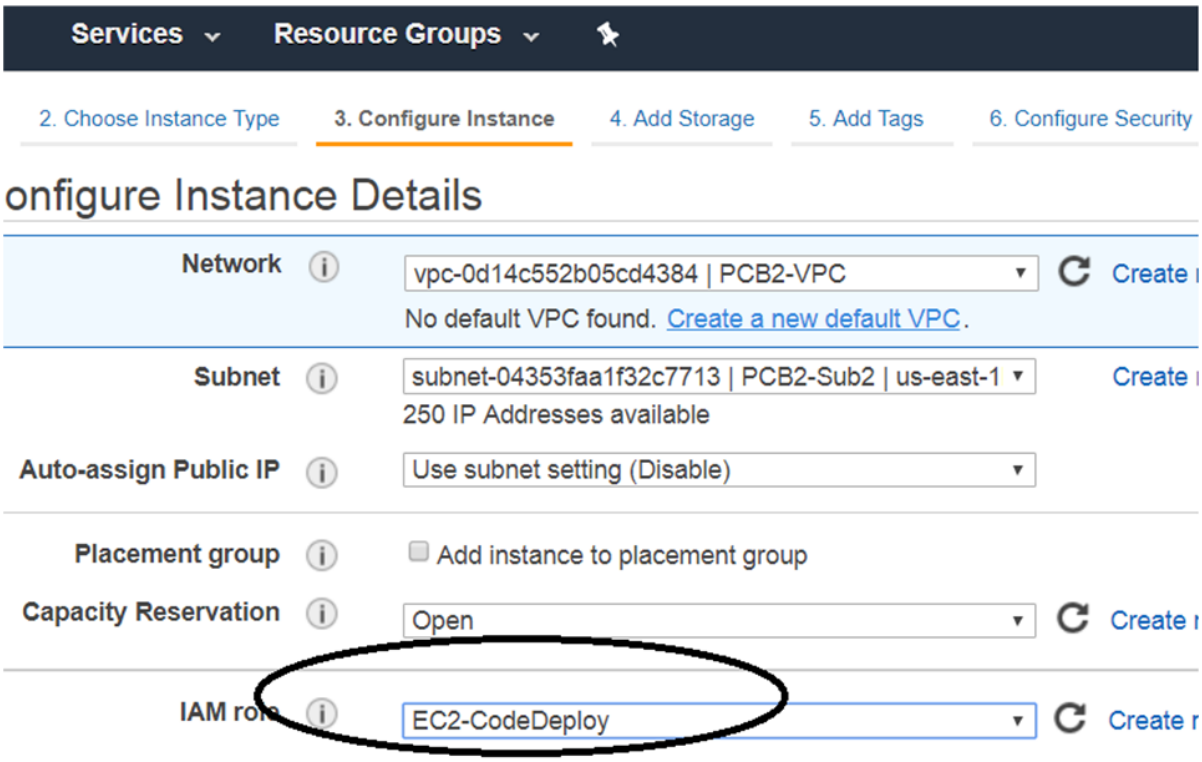
c. IAM Policy -- **Codedeploy** to access EC2 instance.
   Eg:-- "Codedeploy"

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
```

```
                "Sid": "VisualEditor0",
                "Effect": "Allow",
                "Action": [
                    "ec2:*"
                ],
                "Resource": "*"
            }
        ]
}
```

Step 2. Create EC2 instance

Create 2 Instance with the Above IAM role that was created for EC2 "**EC2-codedeploy**"



With Security Group allowing -- SSH and HTTP traffic

Step 3.

Upload the “Sample-linuz.zip” into S3 bucket and give “Public” access.

The bucket needs to have "Versioning" ENABLED.

The file is available in the Google drive
https://drive.google.com/drive/u/1/folders/18wNY2GTIcS3vCufomVo9If5B59P5I8nS

Step 4: Install the Codedeploy agent on the EC2 instance.

#! /bin/bash -- (This is required only if you are automating the agent installation on EC2 page)

```
sudo yum install ruby -y

sudo yum install wget -y

cd /home/ec2-user

wget https://
aws-codedeploy-us-east-1.s3.amazonaws.com/latest/install

chmod +x ./install

sudo ./install auto
```

If you want to automate agent installation while creating EC2

instance.

Under the Advance option copy the above command along with

Resource Kit Bucket Names by Region

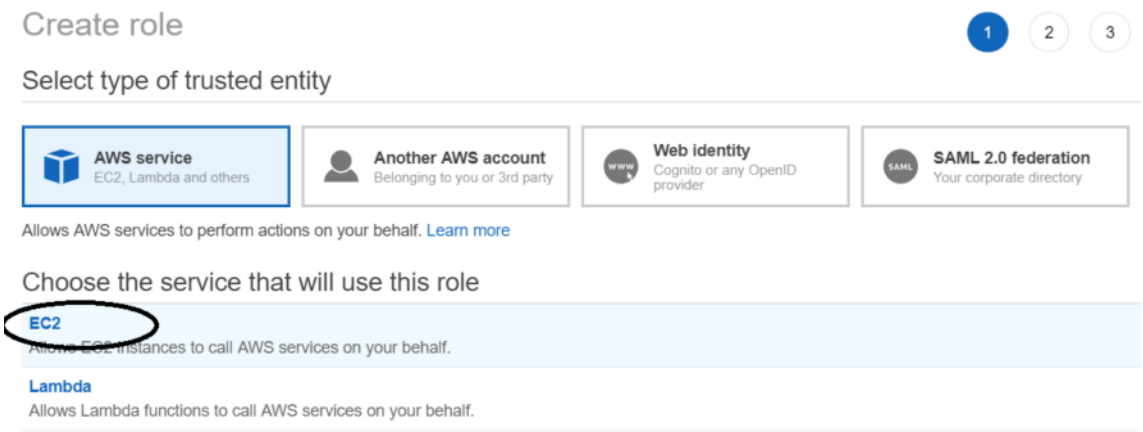
This table lists the names of `bucket-name` replacements required for some procedures in the guide. These are the names of the Amazon S3 buckets that contain the AWS CodeDeploy Resource Kit files.

Region name	`bucket-name` replacement	Region identifier
US East (Ohio)	aws-codedeploy-us-east-2	us-east-2
US East (N. Virginia)	aws-codedeploy-us-east-1	us-east-1
US West (N. California)	aws-codedeploy-us-west-1	us-west-1
US West (Oregon)	aws-codedeploy-us-west-2	us-west-2
Canada (Central)	aws-codedeploy-ca-central-1	ca-central-1
EU (Ireland)	aws-codedeploy-eu-west-1	eu-west-1
EU (London)	aws-codedeploy-eu-west-2	eu-west-2
EU (Paris)	aws-codedeploy-eu-west-3	eu-west-3

EU (Frankfurt)	aws-codedeploy-eu-central-1	eu-central-1
Asia Pacific (Tokyo)	aws-codedeploy-ap-northeast-1	ap-northeast-1
Asia Pacific (Seoul)	aws-codedeploy-ap-northeast-2	ap-northeast-2
Asia Pacific (Singapore)	aws-codedeploy-ap-southeast-1	ap-southeast-1
Asia Pacific (Sydney)	aws-codedeploy-ap-southeast-2	ap-southeast-2
Asia Pacific (Mumbai)	aws-codedeploy-ap-south-1	ap-south-1
South America (São Paulo)	aws-codedeploy-sa-east-1	sa-east-1

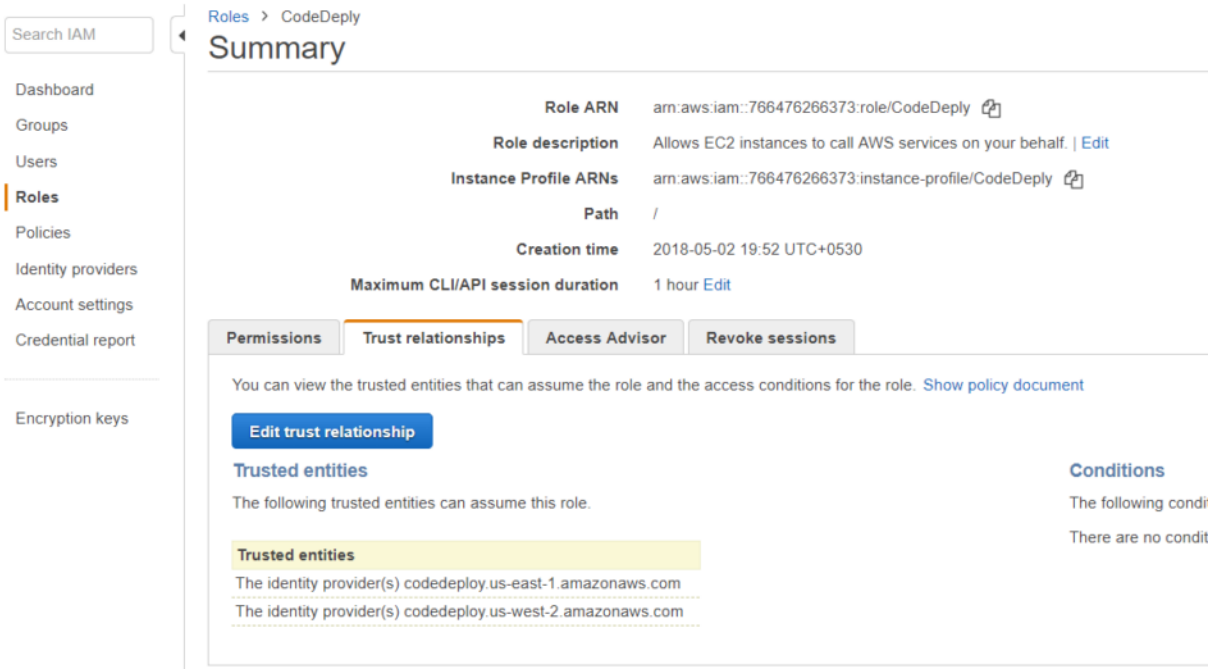
Step 5: Create IAM Role and Policy for **CodeDeploy**

IAM Role for **"CodeDeploy"**



Then attach the above Codedeploy Policy to this IAM role

Add the below trust relationship on the CodeDeploy Role



Click on “Edit trust relationship”

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Sid": "",
            "Effect": "Allow",
            "Principal": {
                "Service": [
                    "codedeploy.us-east-1.amazonaws.com",
                    "codedeploy.us-west-2.amazonaws.com"
                ]
            },
            "Action": "sts:AssumeRole"
        }
    ]
}
```

Step 6: Create an CodeDeploy deployment code.

a.

CodeDeploy helps you to quickly deploy applications to Amazon EC2 instances or on-premises instances.

by creating a deployment that uses a sample application supplied by AWS CodeDeploy, or skip this wizard and
e a custom deployment with your own application.

Sample deployment

Recommended for new AWS CodeDeploy users.

Custom deployment

Recommended if you already have instances and an application to deploy.

Cancel | Skip Walkthrough

AWS CodeDeploy

Create application

Create an application and choose a deployment type. Specify the instances to deploy to. Specify the conditions

Application name* WEBAPP

Compute Platform* EC2/On-premises

Deployment group name* WEBAPPDG01

Deployment type

Choose the deployment to use to deploy your application. Learn more

In-place deployment

Updates the instances in the deployment group with the latest application revision. During
briefly taken offline for its update.

Blue/green deployment

Replaces the instances in the deployment group with new instances and deploys the latest
instances in the replacement environment are registered with a load balancer, instances fr
deregistered and can be terminated.

b. ## Environment configuration

Environment configuration

Specify any combination of Auto Scaling groups, Amazon EC2 instances, and on-premises instances to add instances to this deployment group.

Auto Scaling groups | Amazon EC2 instances | On-premises instances

You can add up to three groups of tags for EC2 instances to this deployment group. Learn more

One tag group : Any instance identified by the tag group will be deployed to.
Multiple tag groups : Only instances identified by all the tag groups will be deployed to.

Tag group 1

	Key	Value	Instances	
1	Name	WS01	2	
2		WS01		

Add tag group

Matching instances

Right now these instances match the criteria you specified. There might be more or fewer instances that match your criteria when a deployment runs.

1 to 2 of 2 instances

Instance ID	Status	Filter types	Association
i-064cb59c2f51d1e66	Running	Name:WS01	EC2
i-0c7b7ceaa6cbddf52	Running	Name:WS01	EC2

c.

Deployment configuration

Choose from a list of default and custom deployment configurations. A deployment configuration is a set of rules that determines how fast an application will be deployed and the success or failure conditions for a deployment.

Deployment configuration CodeDeployDefault.OneAtATime or **Create deployment configuration**

Routes traffic to one instance in the replacement environment at a time. Succeeds if traffic is successfully rerouted to all replacement instances. Fails after the very first rerouting failure. Allows the deployment to succeed for some instances, even if the overall deployment fails.

▸ Advanced

Service role

Select a service role that grants AWS CodeDeploy access to the instances.

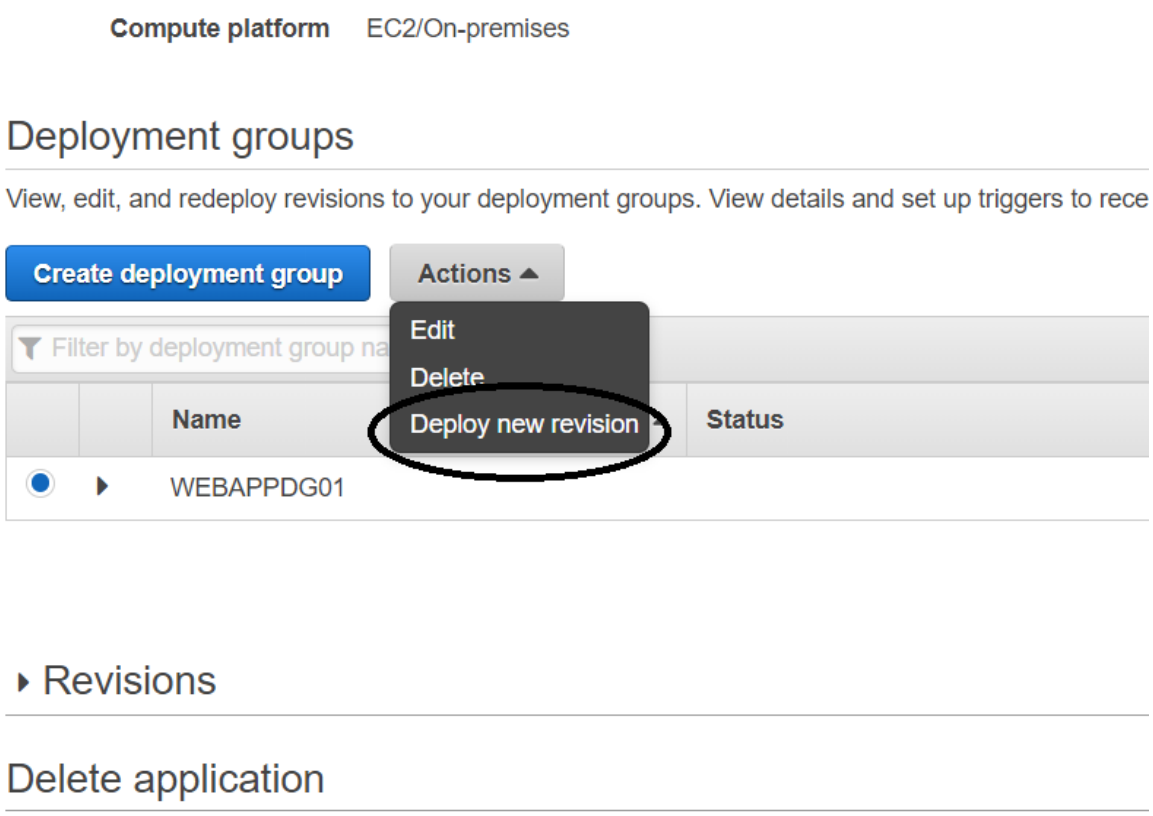
Service role ARN* arn:aws:iam::766476266373:role/CodeDeply

*Required

Cancel | Create application

Successfully Application Deployment Group has been created.

Step 7: Create an 'New Deploy" and point to the S3 location of the Sample-linux.zip file



Click on **"deploy new revision"**on the above screen

Deployment type In-place

Repository type* My application is stored in Amazon S3
My application is stored in GitHub

Revision location* https://s3.amazonaws.com/vishwa19820715/Sampl
Copy and paste the URL of your revision file location from Amazon S3.

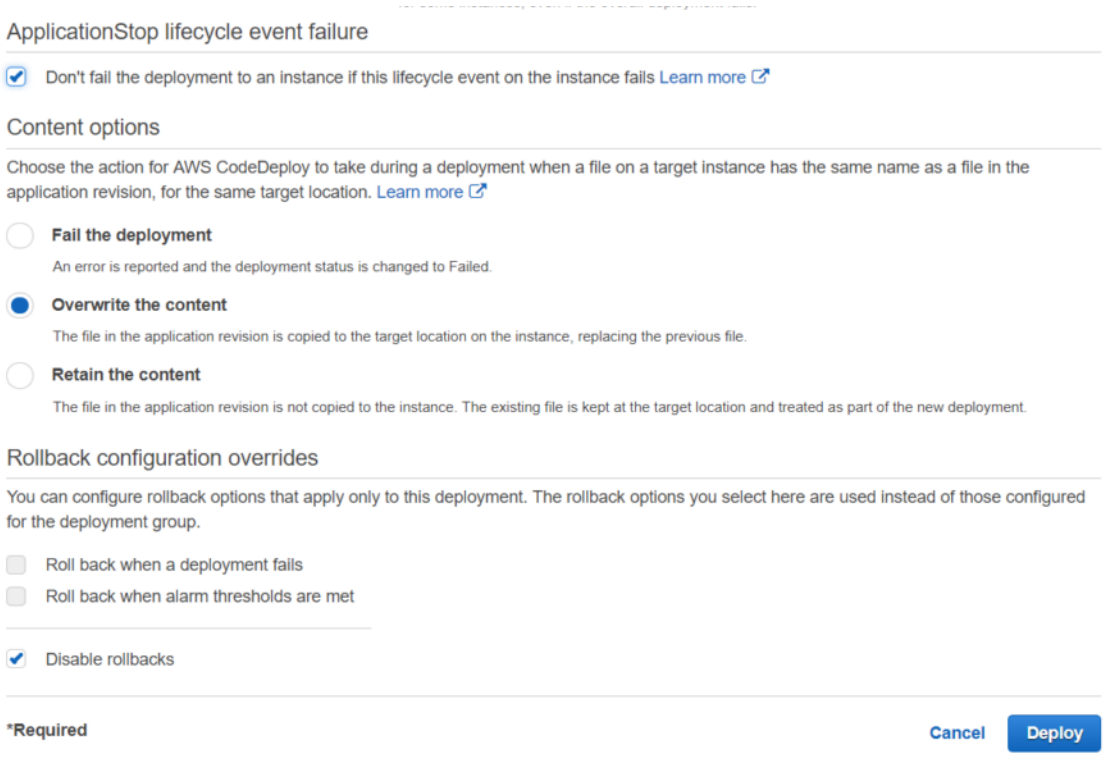
Uploaded to Amazon S3 Sat, 19 May 2018 06:34:32 GMT
The date and time that your revision was uploaded to Amazon S3.

Version ID HfAbZBhxJ1Bx3piWt9v.QtheH.Rk7ARa
The version of the revision to be deployed.

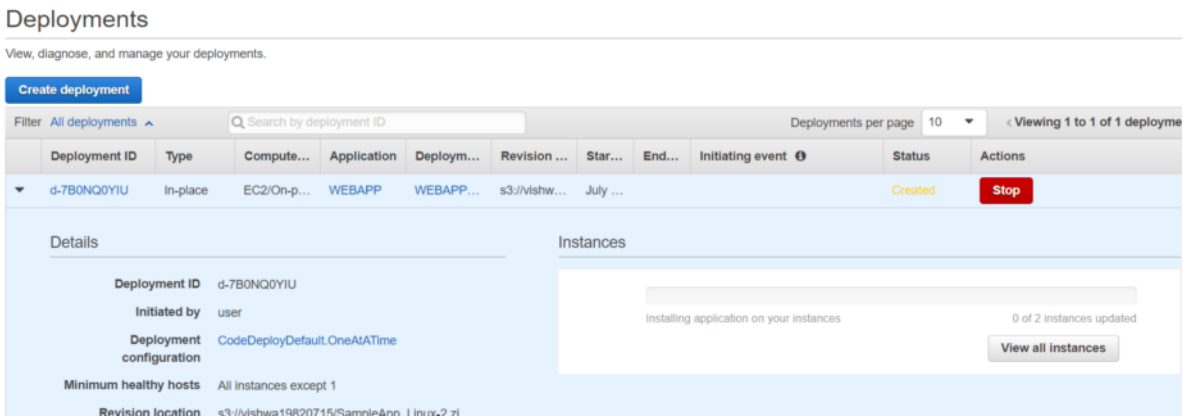
File type* .zip **Supported file type**
The specified file has been detected as an accepted file type (.zip, .tar, .tar.gz).

Deployment description Add a brief description about the deployment. 100 character limit.

Copy the "S3 zip file link" to locate the zip file as above.



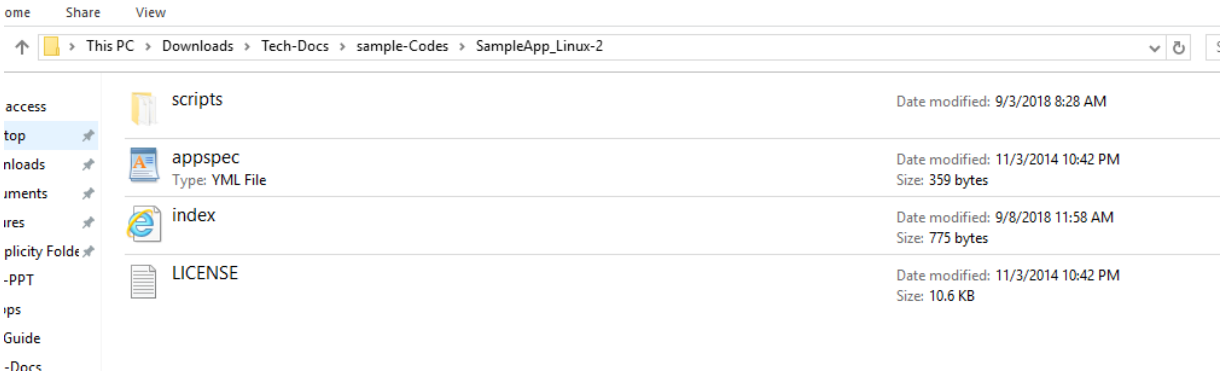
Select "Overwrite the content" if content found in the EC2 instance.



The Above screen would take upto 5 min as there are 2 instance in this example.

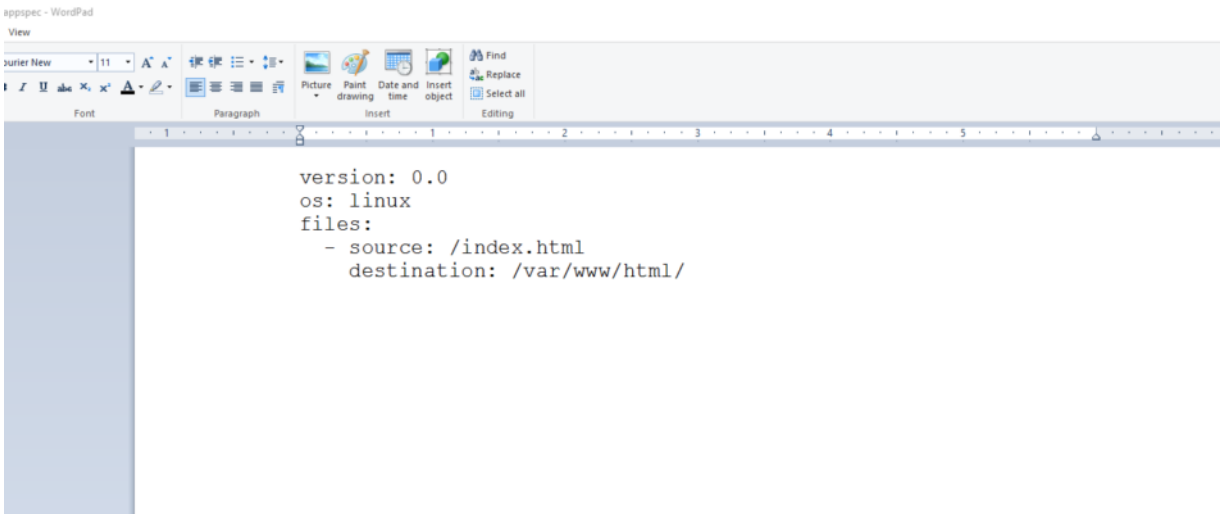
Step 8: Update the Deploy with new content

Download the Zip file on your laptop, uncompress it to a folder.



Remove the “Scripts” folder.

Edit the appspec file as below. Remove all other lines.



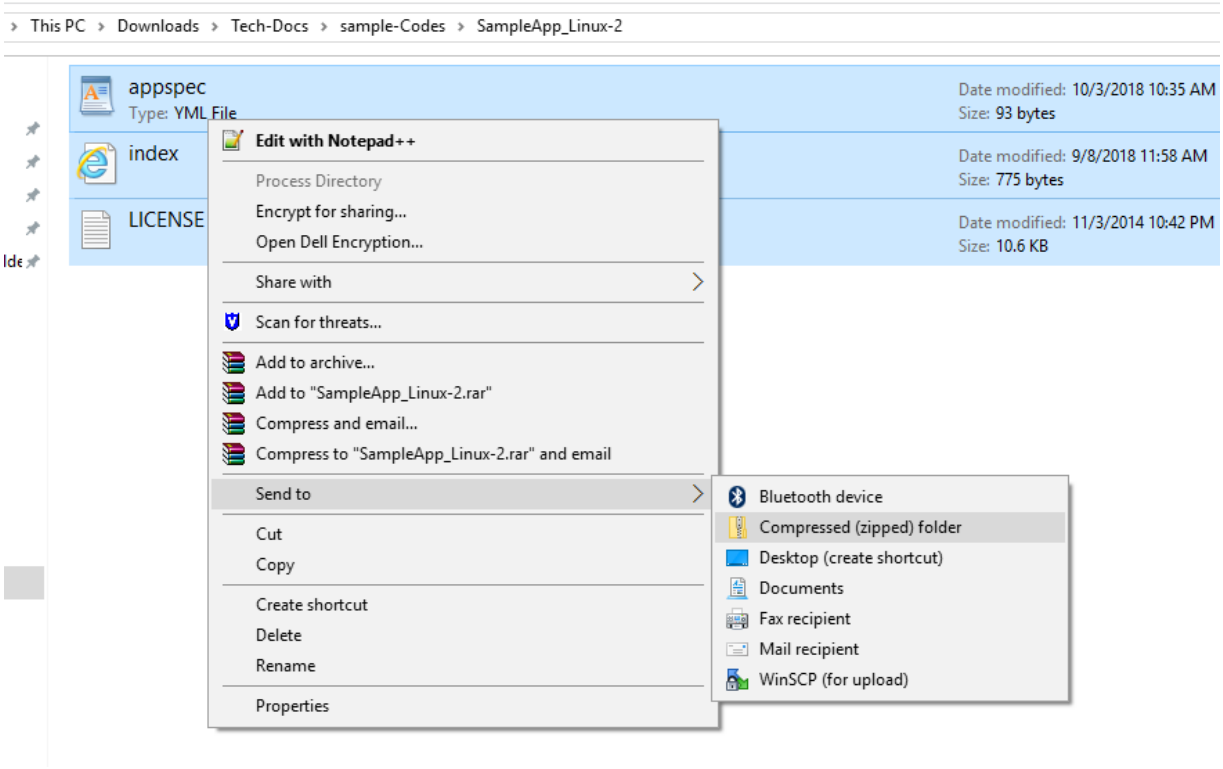
Now edit the “index.html” file. And add some content as below. And save it.

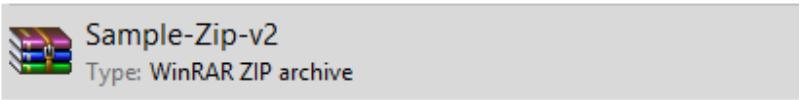
```
index.html
<!DOCTYPE html>
<html>
<head>
  <meta charset="utf-8">
  <title>Sample Deployment</title>
  <style>
    body {
      color: #ffffff;
      background-color: #0188cc;
      font-family: Arial, sans-serif;
      font-size: 14px;
    }

    h1 {
      font-size: 500%;
      font-weight: normal;
      margin-bottom: 0;
    }

    h2 {
      font-size: 200%;
      font-weight: normal;
      margin-bottom: 0;
    }
  </style>
</head>
<body>
  <div align="center">
    <h1>Congratulations Batch, HTTP service is UP</h1>
    <h2>This application was deployed using AWS CodeDeploy. But was customized by Vishwa</
    <p>For next steps, read the <a href="http://aws.amazon.com/documentation/codedeploy">A
    Documentation</a>.</p>
  </div>
</body>
</html>
```

Now zip the file as shown. Select only the FILES and zip and NOT the folder itself, as CODEDEPLOY would be looking for “appsepc” file in the zip parent folder.





Now upload this Sample-zip-v2 to S3, and make it public.

Update the same Deployment Group with “new Revision” with new S3 link.